

Mohamed Shehata

Full-Stack Developer | Backend-Focused (Node.js · NestJS · React · Next.js · TypeScript)

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PROFESSIONAL SUMMARY

Results-driven Full-Stack Developer with 2+ years of hands-on experience architecting and shipping scalable, production-ready web applications. Backend expertise spans Node.js, NestJS, TypeScript, PostgreSQL, MongoDB, and Redis; frontend proficiency covers React.js, Next.js, JavaScript (ES6+), HTML5, CSS3, and Tailwind CSS. Shipped 6+ real-world projects covering enterprise resource planning, e-learning, booking, and social platforms. Proficient with AWS, Docker, CI/CD pipelines, and clean architecture principles. Actively leverages AI tools (LLMs, GitHub Copilot) to accelerate development. Arabic-language technical content creator (Dev-Core) building a growing developer community on YouTube and TikTok.

WORK EXPERIENCE

Full-Stack Developer | Smartly Coded (Software Agency, Egypt)

Nov 2025 – Present

- Engineered a production-grade transportation booking platform (inDriver-style) using React.js and Firebase, handling real-time data for 100+ concurrent users.
- Built real-time trip filtering and live data sync, reducing manual dispatch coordination by approximately 40%.
- Configured Firebase Authentication and Firestore, supporting multi-role permission systems across 3 user types (rider, driver, admin).
- Translated ambiguous client requirements into maintainable features, delivering 4+ major modules on schedule.
- Adopted Next.js and new tooling on the job, cutting initial page load time by 25% via SSR optimization.

Backend Developer | Freelance / Self-Employed

Jun 2024 – Present

- Architected full backend systems for 5+ platforms (ERP, marathon, booking, social-job) using Node.js, NestJS, and TypeScript.
- Enforced OAuth2, JWT authentication, RBAC, and Zod/class-validator API validation, reducing unauthorized access attempts to near zero across all deployments.
- Applied CI/CD pipelines with GitHub Actions, cutting deployment time from a manual ~30 minutes to an automated 5 minutes.
- Containerized services with Docker and Nginx; executed zero-downtime deployments on Railway and Vercel across 4+ projects.

PROJECTS

Inventory & ERP Management System (ERP-Lite) – Full-Stack (NestJS + Next.js)

2026

github.com/Mohamed-Shahata/ERP-Lite

- Built a production-style, multi-module ERP monorepo (NestJS 11 backend, Next.js 16 / React 19 frontend) covering the full procure-to-pay and order-to-cash cycle: Suppliers, Purchase Orders, Sales Orders, Customers, Products, Invoices, Payments, and Stock Movements.
- Designed a strict layered backend architecture (controller/service/repository per domain module) on PostgreSQL with Prisma 7, keeping 14+ business modules independently testable and scalable.
- Implemented secure authentication with httpOnly-cookie JWT access tokens and rotating, hashed refresh tokens (SHA-256), including automatic theft/replay detection and forced re-authentication on password change.
- Enforced defense-in-depth security: Helmet security headers, strict CORS allow-list, global rate limiting (Throttler), a reusable @Sanitize() decorator to strip HTML from free-text fields, and global DTO validation with mass-assignment protection.
- Built a dependency-free, in-memory TTL caching layer (cache-aside pattern) for expensive aggregate queries (Dashboard KPIs, Sales/Purchases/Inventory/Payments Reports), with centralized cache keys and TTLs designed to swap to Redis with a one-file change.
- Delivered a full bilingual (English/Arabic) RTL-ready frontend with role-based routing enforced at the Next.js middleware layer, a reporting suite with date-range filters and PDF/Excel export, an audit log viewer, and admin-editable content pages via a custom Rich Text Editor with server-side HTML sanitization.

Courses Platform – Backend (Node.js + TypeScript)

2025

github.com/Mohamed-Shahata/Courses_platform

- Architected backend for an e-learning platform supporting instructor course creation, student enrollment, progress tracking, and payment integration using Node.js, TypeScript, and PostgreSQL/MongoDB.
- Reduced API response time by approximately 35% via query optimization and Redis caching on high-traffic endpoints.
- Shipped a clean RESTful API with modular architecture, Docker deployment, and an integrated CI/CD pipeline.

Booking Platform – Backend (Node.js + NestJS)

Oct – Nov 2025

github.com/Mohamed-Shahata/booking-platform

- Built backend for a consultation booking platform enabling expert discovery, session booking, and payment processing for 3 service types using Node.js and NestJS.
- Focused on API design, multi-step booking flows, and system scalability using clean architecture patterns.

BITX Marathon – Programming Competition Platform

May – Jun 2025

github.com/Mohamed-Shahata/btix-Backend

- Developed backend for a programming marathon platform where teams register, submit via GitHub, and admins score in real time, supporting 50+ teams at launch.
- Implemented Google/email OAuth2, team management, admin scoring workflow, email notifications, and RBAC with Node.js, TypeScript, and MongoDB.

Aqdam – Social Job Platform

Mar 2025

github.com/Mohamed-Shahata/aqdam_backend

- Created a LinkedIn-style job platform for posting opportunities, profile updates, and listing interactions using NestJS and PostgreSQL.
- Optimized feed query performance by 60% using Redis caching, supporting concurrent reads across 100+ listings.

TECHNICAL SKILLS

- Backend: Node.js, NestJS, Express.js, TypeScript, RESTful APIs, GraphQL, Clean Architecture, OOP
- Frontend: React.js, Next.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Bootstrap
- Databases: PostgreSQL, MongoDB, Redis, Prisma ORM, SQL, Database Design
- Cloud & DevOps: AWS (EC2, S3), Docker, Nginx, CI/CD, GitHub Actions, Railway, Vercel
- Auth & Security: JWT, bcrypt, OAuth2 (Google), RBAC, API Security Best Practices
- OS & Tools: Linux, Git, GitHub, Postman, Swagger, VS Code
- Languages: JavaScript, TypeScript, Python (NumPy, Pandas), C++, Bash
- AI & Tooling: LLMs, GitHub Copilot — applied to accelerate development, debugging, and architecture decisions
- CS Concepts: Data Structures & Algorithms, Software Architecture, Performance Optimization, Clean Code

EDUCATION

General Secondary Certificate (Thanawiya Amma) | Egypt — 2024

Self-taught Full-Stack Developer since graduation (Jun 2024 – Present); continuously shipping real-world projects and deepening expertise in backend systems.

CERTIFICATIONS

- Introduction to Cybersecurity — Cisco Networking Academy (Oct 2025)
- Python – Key Skills — Labble Course (Oct 2025)

LANGUAGES

Arabic (Native) • English (Intermediate)